Yuan, Wenxuan

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https://wenxuan52.github.io/

I am especially interested in research-oriented industry positions at the intersection of world models, self-supervised representation learning, and machine learning for complex dynamical systems, particularly in building models that can predict, reason about, and plan within high-dimensional environments. I am also open to aligned PhD opportunities.

Education Experience

Imperial College London Sep 2024 - Nov 2025

MSc | Applied Computational Science and Engineering | Department of Earth Sciences and Engineering

- ***Degree***: Distinction
- ***Key Courses***: Computational Mathematics (A*), Machine Learning (A*), Fluid Dynamics (A*), Deep Learning (A), Inversion & Optimisation (A) etc.

Taiyuan University of Technology Sep 2020 - Jun 2024

BSc | Mathematics and Applied Mathematics | School of Mathematics

- ***GPA***: 4.5/5.0
- ***Key Courses***: mathematical analysis (97), advanced algebra (98), probability theory and mathematical statistics (93), mathematical physics equations (97), numerical analysis (94), abstract algebra (100) etc.
- ***Honorary Awards***: Outstanding Academic Award (2020-2024), Outstanding Research Award (2022-2024), etc.

Publications

Xiaoyuan Cheng* **Wenxuan Yuan***, Yiming Yang, Yuanzhao Zhang, Sibo Cheng, Yi He, and Zhuo Sun, "*Information Shapes Koopman Representation",* The Fourteenth International Conference on Learning ***(Oral)***, April 2026.

Wenxuan Yuan and Rui Guo, "*Physics-guided multistage neural network: A physically guided network for step initial values and dispersive shock wave phenomena* ", Physical Review E, Volume 110, Issue 6, Page 065307.

Wenxuan Yuan, Shanchuan He, and Jianwen Dou, "*MSTCN-VAE: An unsupervised learning method for micro gesture recognition based on skeleton modality* ", International Joint Conference on Artificial Intelligence ***(Oral)***, August 2023.

Wenxuan Yuan, Rui Guo, and Yining Gao, "*Physics-informed Neural Network method for the Modified Nonlinear Schrödinger equation* ", Optik, Volume 279, Page 170739.

Preprints & Submissions

Xiaoyuan Cheng*, **Wenxuan Yuan***, Boyang Li, Yuanchao Xu, Yiming Yang, Hao Liang, Bei Peng, Robert Loftin, Zhuo Sun, Yukun Hu, "*How Does the Lagrangian Guide Safe Reinforcement Learning through Diffusion Models?*", Submitted to ICML 2026.

Yiming Yang, Xiaoyuan Cheng, Yi He, Kaiyu Li, **Wenxuan Yuan**, Zhuo Sun, "*On Stability and Robustness of Diffusion Posterior Sampling for Bayesian Inverse Problems*", Submitted to ICML 2026.

Competitions

FEMA Predicting the Unpredictable Challenge — Rank 1 (Team Leader) Feb 2025

MiGA-IJCAI Challenge (Track1 Microgesture Classification) — Rank 12 (Team Leader) Apr 2023 - Jun 2023

China Undergraduate Mathematical Contest in Modeling — National Second Prize (Team Leader) Sep 2023

ASC Student Supercomputer Challenge — Global Second Class (Team member) Nov 2021 - Mar 2022

SKILLS

Python、PyTorch、JAX、Git、LaTeX、GPU Parallel